

IBAD ULLAH
25K-6544

Q1:

Code:

```
int triageScore(int age, int severity, int hours){
    int score = (severity * 10) - age + (hours * 2);
    return score;
}

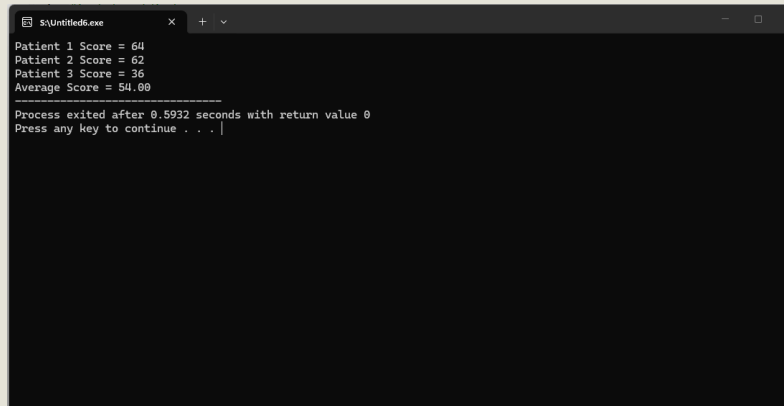
int main(){
    int patients[3][3] = {
        {10,7,2},
        {20,8,1},
        {30,6,3}
    };

    int total = 0;

    for(int i=0;i<3;i++){
        int score = triageScore(patients[i][0], patients[i][1], patients[i][2]);
        printf("Patient %d Score = %d\n", i+1, score);
        total += score;
    }

    printf("Average Score = %.2f", total/3.0);
}
```

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```
5\Untitled6.exe
Patient 1 Score = 64
Patient 2 Score = 62
Patient 3 Score = 36
Average Score = 54.00
-----
Process exited after 0.5932 seconds with return value 0
Press any key to continue . . . |
```

Q2:

Code:

```
#include <stdio.h>
```

```
float applyDiscount(float price, int member){
    if(member) return price * 0.8;
    else return price * 0.9;
}
```

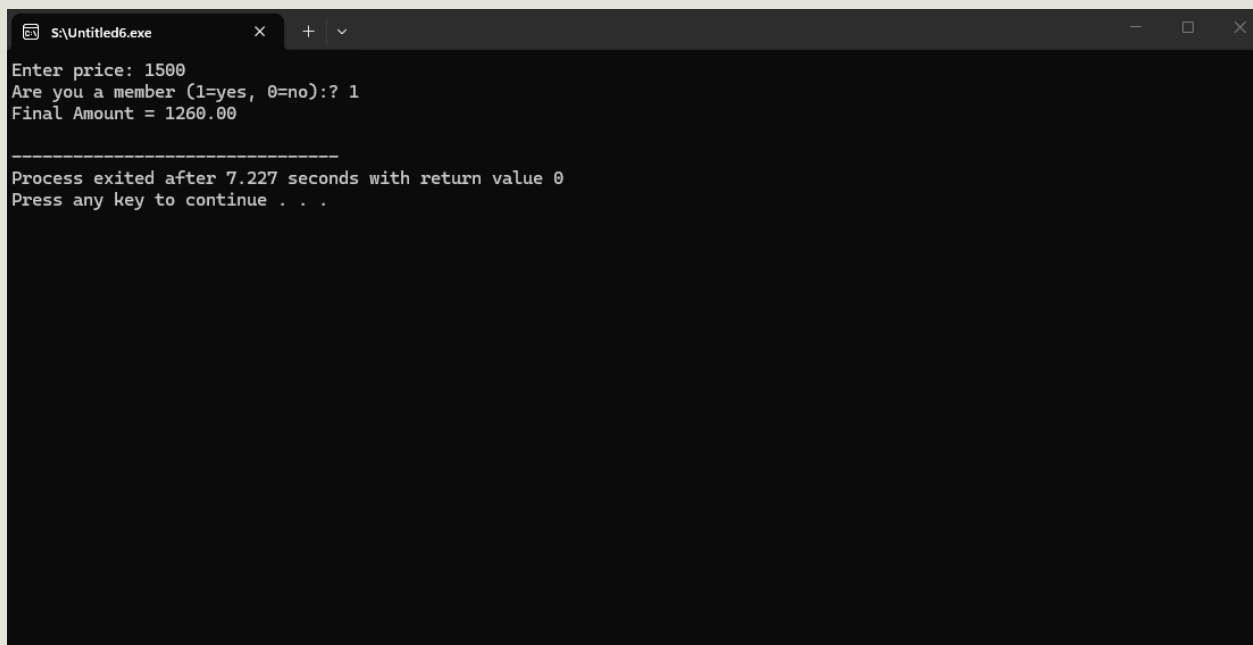
```
float applyTax(float price){
    return price * 1.05;
}
```

```
int main(){
    float price;
    int member;

    printf("Enter price: ");
    scanf("%f", &price);
```

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```
printf("Are you a member (1=yes, 0=no):? ");  
scanf("%d", &member);  
  
float final = applyTax(applyDiscount(price, member));  
printf("Final Amount = %.2f\n", final);  
}  
printInvoice(finalAmount);  
}
```



```
S\Untitled6.exe  
Enter price: 1500  
Are you a member (1=yes, 0=no):? 1  
Final Amount = 1260.00  
-----  
Process exited after 7.227 seconds with return value 0  
Press any key to continue . . .
```

Q3:

Code:

```
#include <stdio.h>
```

```
int totalRuns(int r[]){  
    int sum=0;  
    for(int i=0;i<10;i++) sum+=r[i];
```

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```
    return sum;  
}
```

```
int highest(int r[]){  
    int max=r[0];  
    for(int i=1;i<10;i++)  
        if(r[i]>max) max=r[i];  
    return max;  
}
```

```
int above30(int r[]){  
    int count=0;  
    for(int i=0;i<10;i++)  
        if(r[i]>30) count++;  
    return count;  
}
```

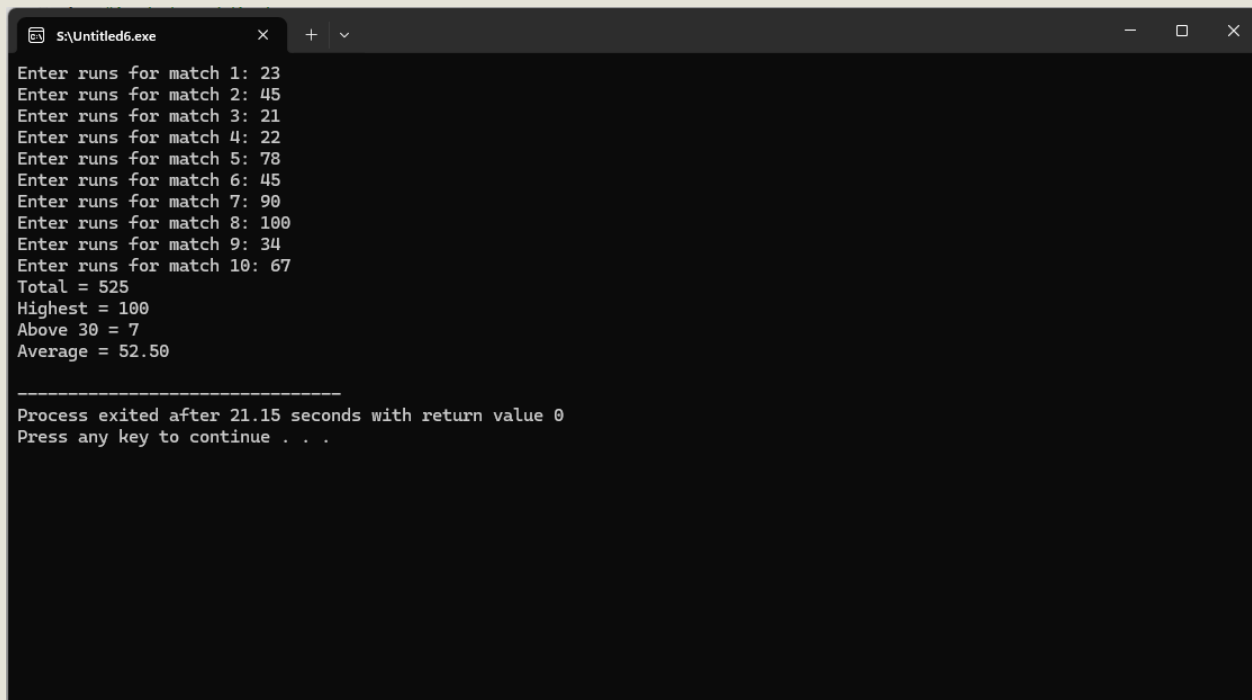
```
float average(int r[]){  
    return totalRuns(r)/10.0;  
}
```

```
int main(){  
    int runs[10];  
  
    for(int i=0;i<10;i++){  
        printf("Enter runs for match %d: ", i+1);  
        scanf("%d",&runs[i]);  
    }
```

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```
}

printf("Total = %d\n", totalRuns(runs));
printf("Highest = %d\n", highest(runs));
printf("Above 30 = %d\n", above30(runs));
printf("Average = %.2f\n", average(runs));
}
```



```
S\Untitled6.exe
Enter runs for match 1: 23
Enter runs for match 2: 45
Enter runs for match 3: 21
Enter runs for match 4: 22
Enter runs for match 5: 78
Enter runs for match 6: 45
Enter runs for match 7: 90
Enter runs for match 8: 100
Enter runs for match 9: 34
Enter runs for match 10: 67
Total = 525
Highest = 100
Above 30 = 7
Average = 52.50

-----
Process exited after 21.15 seconds with return value 0
Press any key to continue . . .
```

Q4:

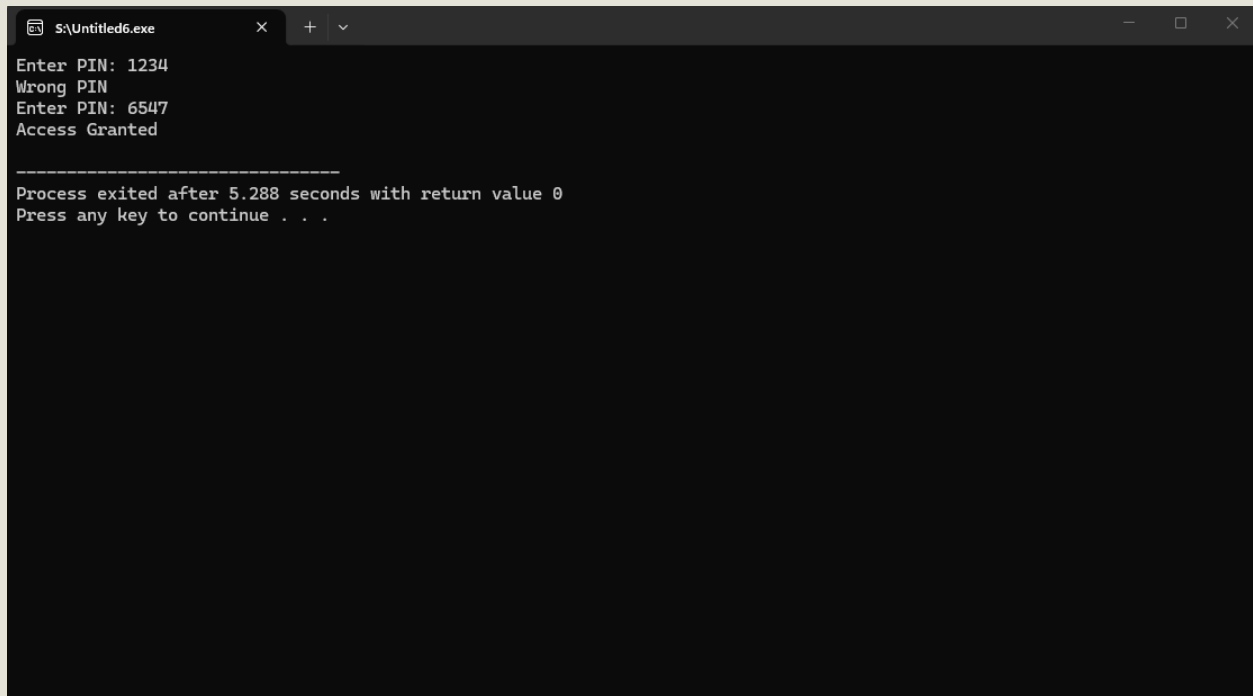
Code:

```
int isValidPIN(int entered, int correct){
    return entered == correct;
}
```

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```
int main(){  
    int correctPIN = 4729, pin;  
  
    for(int i=1;i<=3;i++){  
        printf("Enter PIN: ");  
        scanf("%d",&pin);  
  
        if(isValidPIN(pin, correctPIN)){  
            printf("Access Granted\n");  
            return 0;  
        } else {  
            printf("Wrong PIN\n");  
        }  
    }  
    printf("Card Blocked\n");  
}
```

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```
S:\Untitled6.exe
Enter PIN: 1234
Wrong PIN
Enter PIN: 6547
Access Granted

=====
Process exited after 5.288 seconds with return value 0
Press any key to continue . . .
```

Q5:

Code:

```
float megajoulesToKiloW(float m){ return m * 1000; }
```

```
float kiloWToMilli(float k){ return k * 1000; }
```

```
int main(){
```

```
    float megajoules;
```

```
    printf("Enter value in Mega: ");
```

```
    scanf("%f",&megajoules);
```

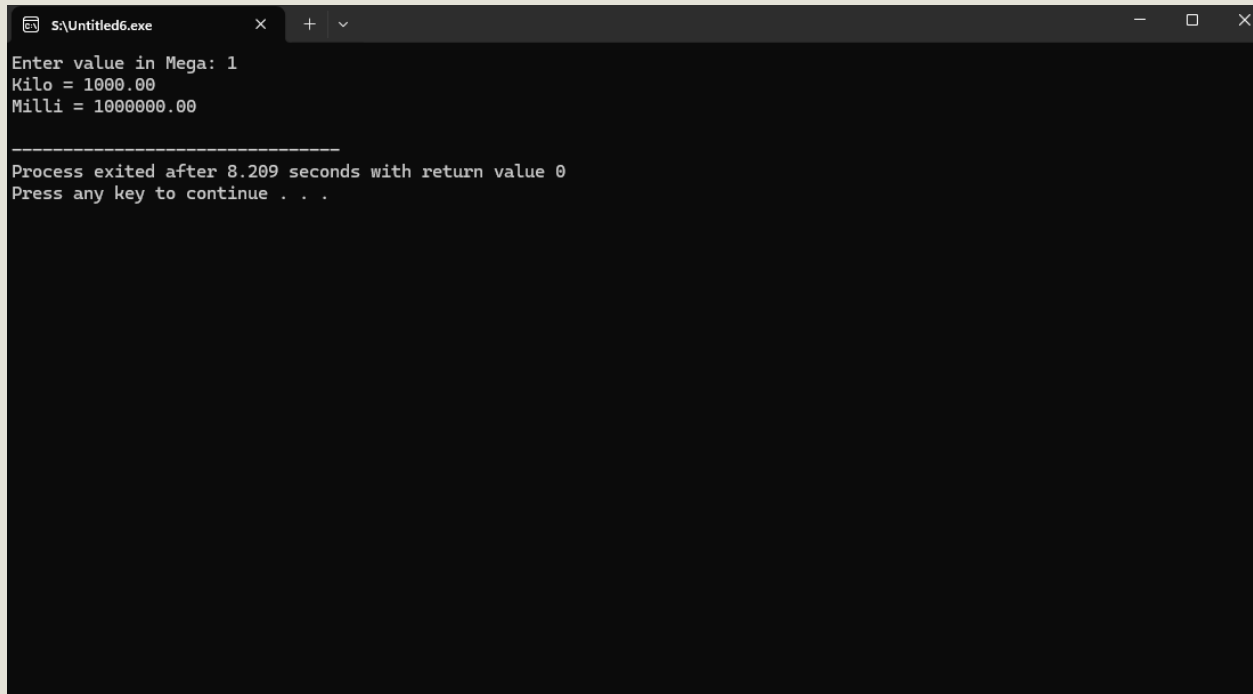
```
    float kiloW = megajoulesToKiloW(megajoules);
```

```
    float milli = kiloWToMilli(kiloW);
```

```
    printf("Kilo = %.2f\n", kiloW);
```

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```
printf("Milli = %.2f\n", milli);  
}
```



```
S:\Untitled6.exe  
Enter value in Mega: 1  
Kilo = 1000.00  
Milli = 1000000.00  
-----  
Process exited after 8.209 seconds with return value 0  
Press any key to continue . . .
```

Q6:

Code:

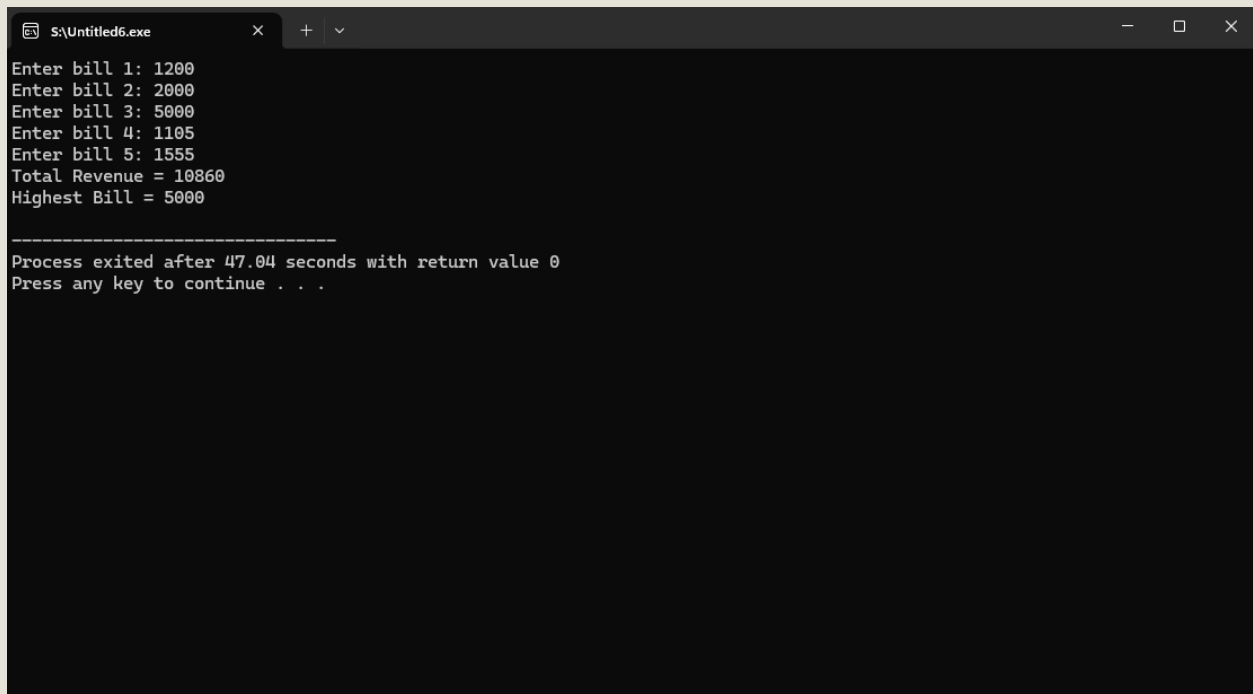
```
#include <stdio.h>
```

```
int total(int b[]){  
    int sum=0;  
    for(int i=0;i<5;i++) sum+=b[i];  
    return sum;  
}
```

```
int highest(int b[]){  
    int max=b[0];  
    for(int i=1;i<5;i++)
```


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```
        if(b[i]>max) max=b[i];  
    return max;  
}  
  
int main(){  
    int bills[5];  
  
    for(int i=0;i<5;i++){  
        printf("Enter bill %d: ", i+1);  
        scanf("%d",&bills[i]);  
    }  
  
    printf("Total Revenue = %d\n", total(bills));  
    printf("Highest Bill = %d\n", highest(bills));  
}
```



```
S:\Untitled6.exe  
Enter bill 1: 1200  
Enter bill 2: 2000  
Enter bill 3: 5000  
Enter bill 4: 1105  
Enter bill 5: 1555  
Total Revenue = 10860  
Highest Bill = 5000  
  
-----  
Process exited after 47.04 seconds with return value 0  
Press any key to continue . . .
```

Q7:

```
#include <stdio.h>
```

```
void scale(int m[], int size){
```

```
    int max=m[0];
```

```
    for(int i=1;i<size;i++)
```

```
        if(m[i]>max) max=m[i];
```

```
    for(int i=0;i<size;i++)
```

```
        m[i] = (m[i]*100)/max;
```

```
}
```

```
int main(){
```

```
    int marks[5];
```

```
    for(int i=0;i<5;i++){
```

```
        printf("Enter marks %d: ", i+1);
```

```
        scanf("%d",&marks[i]);
```

```
    }
```

```
    scale(marks,5);
```

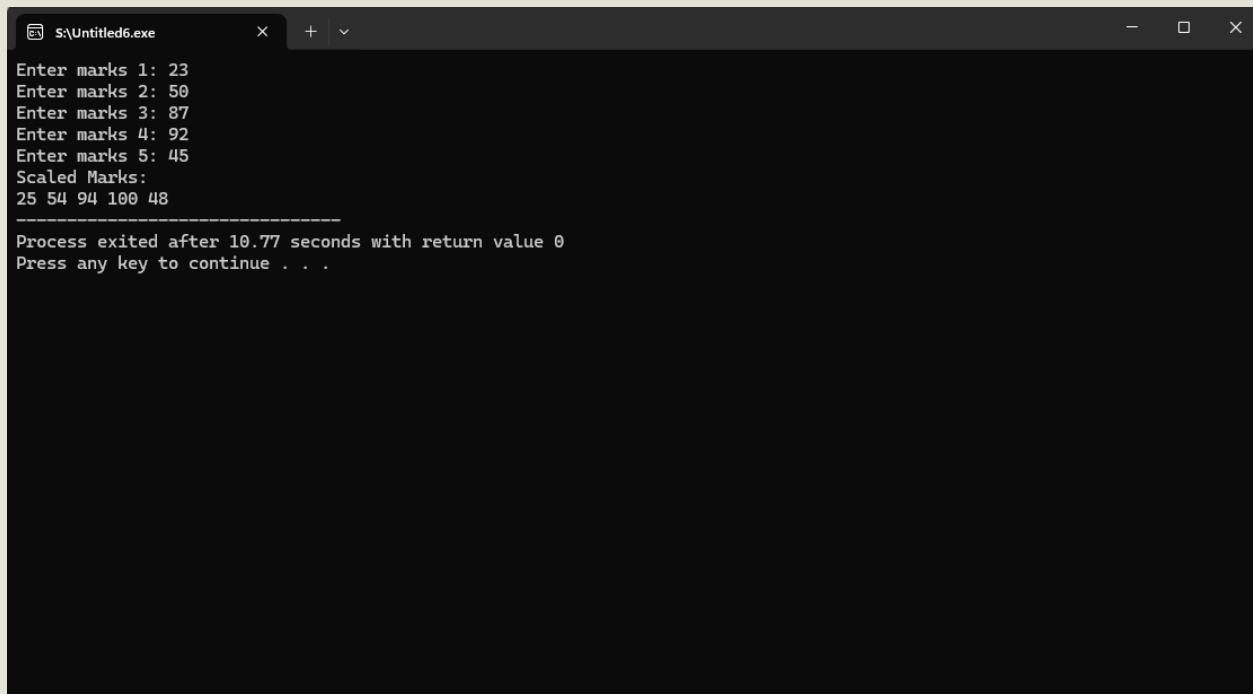
```
    printf("Scaled Marks:\n");
```

```
    for(int i=0;i<5;i++)
```

```
        printf("%d ", marks[i]);
```

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}



```
S:\Untitled6.exe
Enter marks 1: 23
Enter marks 2: 50
Enter marks 3: 87
Enter marks 4: 92
Enter marks 5: 45
Scaled Marks:
25 54 94 100 48
=====
Process exited after 10.77 seconds with return value 0
Press any key to continue . . .
```

Q8:

Code:

```
void printData(int vib, float temp, char grade){
    printf("Vibration %d\n", vib);
    printf("temperature= %.2f\n",temp);
    printf("grade=%c \n", grade);
}
```

```
int main(){
    int vib;
    float temp;
    char grade;
```

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```
printf("Enter vibration: ");
```

```
scanf("%d",&vib);
```

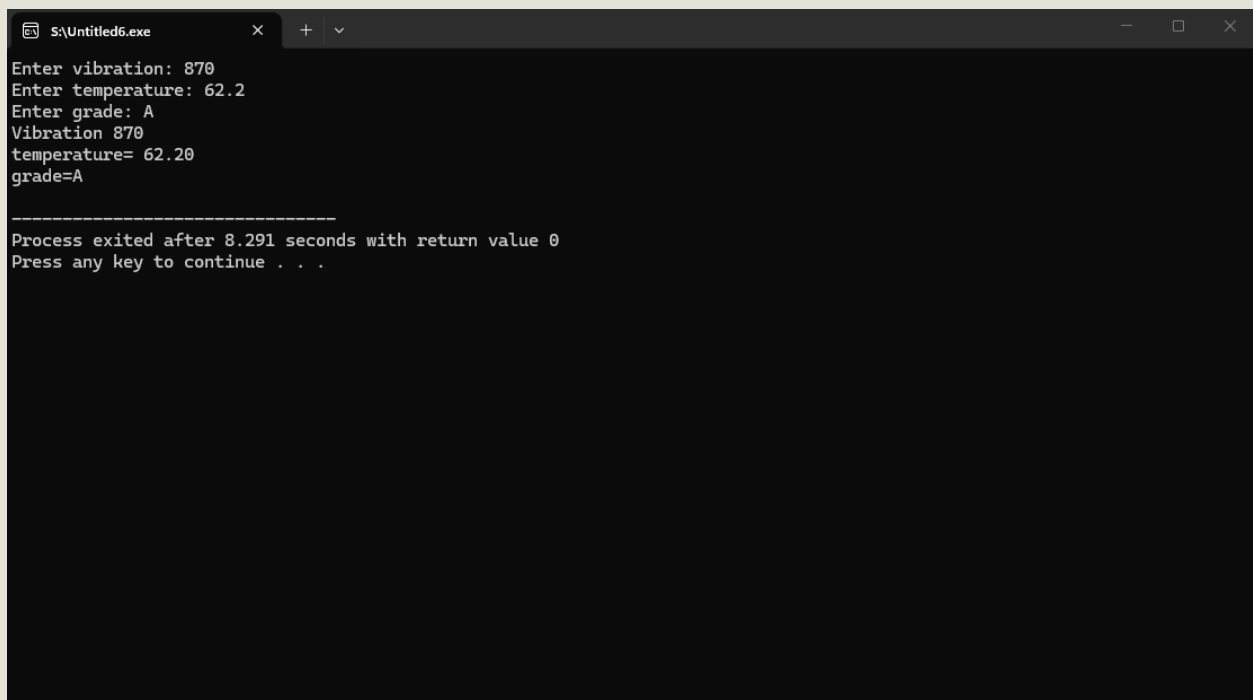
```
printf("Enter temperature: ");
```

```
scanf("%f",&temp);
```

```
printf("Enter grade: ");
```

```
scanf(" %c",&grade);
```

```
printData(vib,temp,grade);
```



```
S:\Untitled6.exe
Enter vibration: 870
Enter temperature: 62.2
Enter grade: A
Vibration 870
temperature= 62.20
grade=A

-----
Process exited after 8.291 seconds with return value 0
Press any key to continue . . .
```

Q9:

Code:

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```
void analyze(int a[4][7]){
    for(int i=0;i<3;i++){
        int sum=0;
        for(int j=0;j<7;j++){
            printf("%d ", a[i][j]);
            sum+=a[i][j];
        }
        printf(" student attendance= %d\n", sum);
    }
}
```

```
int main(){
    int att[4][7];

    for(int i=0;i<3;i++){
        for(int j=0;j<7;j++){
            printf("Student %d Day %d: ", i+1, j+1);
            scanf("%d",&att[i][j]);
        }
    }

    analyze(att);
}
```

```
S:\Untitled6.exe
Student 1 Day 1: 1
Student 1 Day 2: 1
Student 1 Day 3: 1
Student 1 Day 4: 1
Student 1 Day 5: 1
Student 1 Day 6: 1
Student 1 Day 7: 1
Student 2 Day 1: 0
Student 2 Day 2: 0
Student 2 Day 3: 0
Student 2 Day 4: 0
Student 2 Day 5: 1
Student 2 Day 6: 1
Student 2 Day 7: 1
Student 3 Day 1: 0
Student 3 Day 2: 1
Student 3 Day 3: 0
Student 3 Day 4: 1
Student 3 Day 5: 0
Student 3 Day 6: 1
Student 3 Day 7: 0
1 1 1 1 1 1 student attendance= 7
0 0 0 0 1 1 1 student attendance= 3
0 1 0 1 0 1 0 student attendance= 3

-----
Process exited after 16.95 seconds with return value 0
Press any key to continue . . . |
```

Q10:

Code:

```
#include <stdio.h>
```

```
void basic(int dmg, int hp){
```

```
    printf("Basic Attack! Enemy HP = %d\n", hp-dmg);
```

```
}
```

```
void power(int dmg, int hp){
```

```
    printf("Power Strike! Enemy HP = %d\n", hp-(2*dmg));
```

```
}
```

```
void heal(int hp){
```

```
    printf("Heal! HP = %d\n", hp+20);
```

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```
}
```

```
void dispatch(int action, int dmg, int hp){
```

```
    if(action==1) basic(dmg, hp);
```

```
    else if(action==2) power(dmg, hp);
```

```
    else if(action==3) heal(hp);
```

```
    else printf("Invalid Action\n");
```

```
}
```

```
int main(){
```

```
    int action, dmg, hp;
```

```
    printf("Enter action (1=basic,2=power,3=heal): ");
```

```
    scanf("%d",&action);
```

```
    printf("Enter damage: ");
```

```
    scanf("%d",&dmg);
```

```
    printf("Enter HP: ");
```

```
    scanf("%d",&hp);
```

```
    dispatch(action, dmg, hp);
```

```
}
```

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```
S:\Untitled6.exe x + v
Enter action (1=basic,2=power,3=heal): 1
Enter damage: 50
Enter HP: 100
Basic Attack! Enemy HP = 50

-----
Process exited after 9.558 seconds with return value 0
Press any key to continue . . .
```